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A Novel Design to Reduce Uncertainty in Remaining Oil Saturation Estimation Through Integrated Laboratory Core Measurements: A Case Study from Kuwait's Heterogeneous Oolitic Carbonate Reservoir

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Abstract

This study aims to reduce uncertainty in remaining oil saturation (ROS) estimation in Kuwait's heterogeneous Minagish Oolite carbonate reservoir. A progressive experimental workflow across three batches integrates Liquid Trapper coring, Dean-Stark, Core flooding, Centrifuge and NMR (T1/T2) techniques. Objectives include improving material balance, distinguishing hydrocarbon from water signals, and validating ROS and Sor using multiple methods. Deuterium tracer and doped brine enhance saturation accuracy. Results guide EOR efficiency and recovery factor estimation.

A progressive stepwise laboratory workflow was designed to quantify ROS using cores acquired via Liquid Trapper coring. Learnings from every batch experiments were used for improvements in later phases. The workflow incorporated routine core analysis, Dean- Stark extraction, core flooding, centrifuge testing, and NMR including native-state and doped-brine measurements. Deuterium tracers assessed mud invasion, while doped brine enhanced hydrocarbon-water signal separation. Batch-1 showed fluid loss during core retrieval. In response, batches 2 and 3 integrated improved NMR protocols and centrifuge methods. These refinements enhanced material balance, reduced uncertainty in ROS estimation, and enabled reliable saturation measurements in swept and unswept zones.

A Batchwise experimental workflow adopted to evaluate ROS in cores obtained by Liquid Trapper coring helped in optimising core sample selection process, enhancing cost economics and time efficiency in laboratory experiments. Batch-1 results indicated unreliable ROS estimates due to hydrocarbon losses possibly during core retrieval and handling. RCA and Dean-Stark analyses yielded material imbalance ($So+Sw < 0.7$), though trends aligned with log data. To address this, Batch-2 incorporated NMR (native, two-phase, single-phase) and core flooding, improving data quality and achieving better material balance ($So+Sw > 0.95$). However, some plugs still failed to yield oil during flooding, especially in partially swept zones. Batch-3 further enhanced the workflow by integrating centrifuge and doped-brine NMR methods. This allowed clearer hydrocarbon-water phase separation, confirmed by distinct NMR T2 signals. Doped-brine NMR improved hydrocarbon quantification and validated helium and saturation porosity

results. Overall, the multi-method approach, including NMR and deuterium tracers, significantly reduced uncertainty in ROS estimation, revealed losses likely occurred during coring/lifting, and improved the confidence in oil saturation measurements—especially in complex, partially swept zones. The study confirms that a combined use of core flooding, Dean-Stark, NMR, and centrifuge methods is essential for accurate Sor/ROS evaluation in such reservoirs.

This study advances the state of knowledge by demonstrating a robust, multi-technique workflow for accurately measuring Remaining Oil Saturation (ROS) in complex carbonate reservoirs. By combining core flooding, NMR (including doped-brine), centrifuge and Dean- Stark methods, it addresses fluid loss challenges during coring and improves saturation estimates. This enhances the reliability of EOR project evaluations, guides better reservoir management, and sets a benchmark for ROS analysis in heterogeneous, partially swept reservoirs.

Introduction and objectives

EOR pilot projects play a critical role in determining the commercial viability of EOR methods for specific reservoirs. These pilots are typically small-scale tests often covering less than 5 acres designed to evaluate the effectiveness of a selected EOR technique under field conditions. In alignment with Kuwait Oil Company's (KOC) 2040 strategy, screening studies were conducted across multiple reservoirs in Kuwait to identify suitable Enhanced Oil Recovery (EOR) methods. As a result, a miscible gas EOR pilot was implemented in the heterogeneous oolitic carbonate reservoir of the Minagish Field in Kuwait. The pilot employed an inverted four-spot pattern, consisting of one central gas injector, three producers, and two observation wells. Extensive surveillance activities were conducted throughout the pilot to gather data on gas flood front movement, inflow profiles, and sweep efficiency (Satish et al., SPE 2024; Pradhan et al., SPE 2023; Al-Mayyan et al., SPE 2023; Al-Murayri et al., SPE 2018).

One of the key objectives of the pilot was to estimate oil recovery following hydrocarbon gas injection into targeted reservoir zones, classified using log data into:

- Unwept zones (Sw near connate water saturation),
- Partially swept zones (Sw near remaining oil saturation), and
- Completely swept zones (Sw near residual oil saturation after waterflooding).

Given the presence of both partially and fully swept zones, it was essential to accurately determine the remaining oil saturation (ROS) within the pilot area, while minimizing uncertainty. Initial ROS estimates were derived from saturation interpretations of open-hole logs across all pilot wells. To validate these log-derived saturations, laboratory core measurements were used. However, core-derived ROS values often suffer from uncertainty due to potential fluid loss during core retrieval and handling.

In the first pilot well, conventional core samples were acquired from the target reservoir zone. Routine Core Analysis (RCA) and Dean- Stark extractions were performed, but the resulting ROS values did not align with the log-based estimates in partially and fully swept zones raising concerns over data reliability. Learning from this, a more rigorous coring approach was adopted for the second pilot well. The primary objective was to trap all fluids released during coring and transportation, to obtain representative fluid saturations. The secondary objective was to assess and minimize the impact of mud invasion by introducing a deuterium tracer into the mud system. To achieve both goals, the Liquid Trapper coring method available commercially was employed.

This paper presents findings from a study aimed at reducing uncertainty in ROS estimation for Kuwait's complex Minagish Oolite reservoir. A structured experimental workflow was implemented over three batches using Liquid Trapper coring, with integrated laboratory analyses including Dean-Stark, core flooding, centrifuge, and NMR (T1/T2). The study focused on:

1. Improving material balance,
2. Differentiating hydrocarbon and water signals,
3. Validating ROS and Sor (residual oil saturation) using multiple methods.

It also highlights the effectiveness of using deuterium tracers in the mud and doped brine during NMR analysis to enhance saturation accuracy. Overall, the study provides valuable insights for optimizing EOR strategies and improving recovery factor estimation in complex carbonate reservoirs.

Reservoir description of the Minagish Formation

The Minagish Formation, dated to the Valanginian stage of the Lower Cretaceous (~121-128 million years), is a ramp carbonate sequence bounded by the Makhul Formation below and the Ratwai Formation above. It is subdivided into upper, middle, and lower units:

- The Upper and Lower Minagish units mainly consist of mudstone and packstone facies.
- The Middle Minagish is the primary hydrocarbon-bearing interval, featuring oolitic grainstones (sublayers 1-9) and bindstone/bafflestone (sublayers 11-13).

Significant diagenetic alteration during subaerial exposure has resulted in the development of weathered zones and considerable heterogeneity. While matrix porosity remains relatively consistent within a zone, permeability varies widely due to interconnected vugs, particularly in weathered intervals (Satish et al., SPE 2024; Al-Saqran et al., SPE 2012).

Water saturation distribution across the reservoir (Fig. 1) clearly illustrates this heterogeneity. Vertically, the reservoir transitions from fully swept zones (water-dominated) at the base to partially swept and unswept zones (oil-dominated) toward the top. The Remaining Oil Saturation (ROS) in these zones is a critical parameter influencing the success of gas EOR and must be accurately estimated using both log and core data. In contrast, Residual Oil Saturation (Sor) in the fully swept zones represents immobile oil. Quantifying Sor is equally important, as it helps assess the fraction of ROS that remains mobile and supports calculation of recovery factor, sweep efficiency, and displacement efficiency after miscible gas injection. Additionally, Sor values aid in refining and recalibrating investment strategies for other EOR techniques.

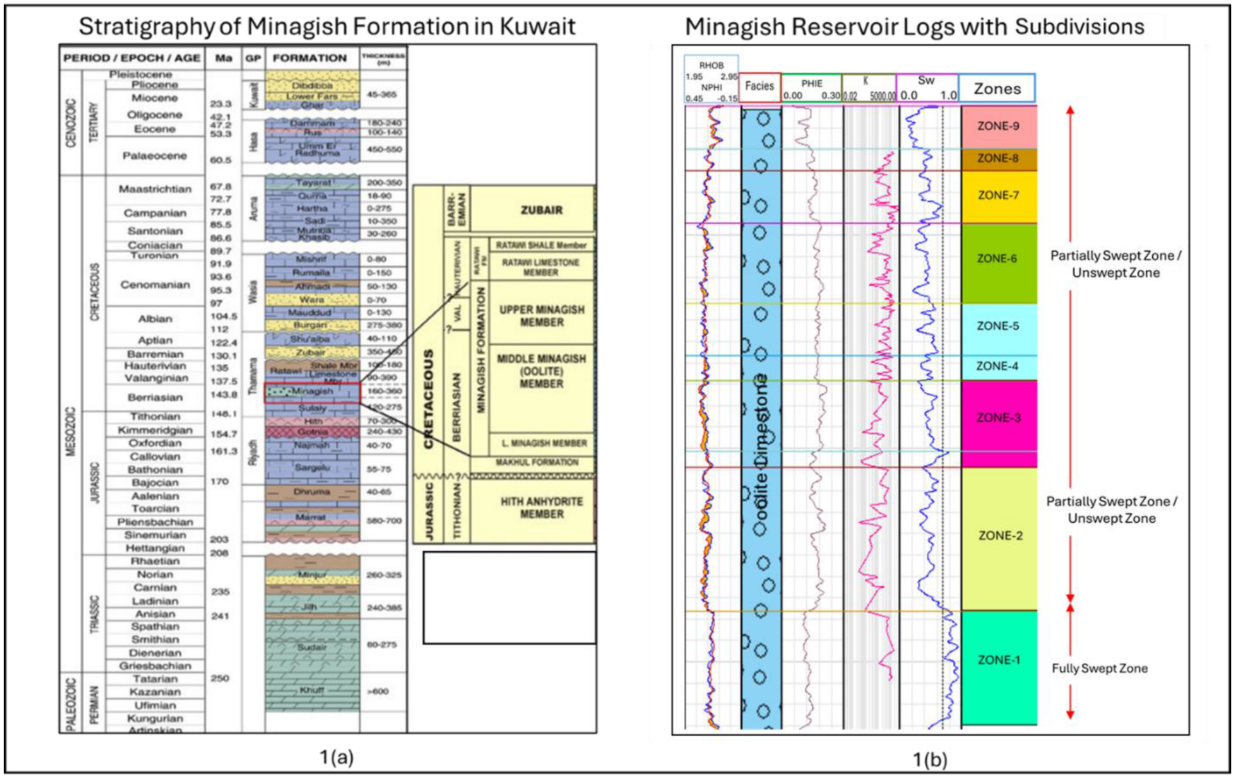


Figure 1—(a) Stratigraphic framework of the Minagish Formation in Kuwait, highlighting the upper, middle, and lower members. 1(b) Representative log section delineating the internal sub-zones of the Minagish reservoir. Core plugs from Liquid Trapper cores for laboratory Remaining Oil Saturation (ROS) estimation was obtained from Zones 1 to 5, which represent the primary study intervals. Based on log-derived saturation profiles, the reservoir is further classified into fully swept and partially swept zones (indicated by red arrows)

Coring in pilot wells

Coring is essential for determining key reservoir parameters such as porosity, permeability, and saturation. While porosity and permeability are inherent and stable under reservoir conditions, oil and water saturations vary due to production and injection. Accurate saturation measurements are crucial for EOR success, especially for estimating **Remaining Oil Saturation (ROS)** as illustrated in the vertical sweep profile (Fig. 1).

Conventional coring often fails to preserve in-situ saturations due to pressure drop during core retrieval, which causes gas to come out of solution and displace fluids from the core. Standard core barrels lack mechanisms to trap expelled fluids, leading to saturation measurement errors. To overcome this limitation, Liquid Trapper coring technology was used in the EOR pilot to retain core fluids and reduce uncertainty in ROS estimation.

To optimize cost and data quality, conventional coring was performed in the upper reservoir layers while Liquid Trapper coring was used in the lower, EOR-targeted zones. One observation well strategically located between the injector and producer pilot wells along the expected gas flood path was cored. Although the primary objective of the liquid trapper core was to measure fluid saturations with reasonable accuracy, it is also used with two existing conventional cores from pilot wells to enhance stratigraphic correlation, particularly in understanding baffle extensions within the reservoir.

Understanding the Liquid Trapper coring system is essential for designing accurate laboratory experiments to estimate Remaining Oil Saturation (ROS). The Liquid Trapper system consists of a modular inner tube made of 1-meter-long sealed cells. These cells are designed to trap expelled fluids in the annular space between the core and the inner wall during core retrieval. An additional safety feature of the Liquid Trapper system is its gas-venting mechanism, which allows liberated gases to escape safely into the mud system during core retrieval. This prevents gas entrapment and reduces the risk of over-pressurization during trip-out operations.

Before deployment, the system is pre-saturated with a selected fluid, tailored to the analytical objectives. In our study, the fluid was

doped with Deuterium oxide (D_2O), a stable isotope tracer, to help detect and quantify mud filtrate invasion during coring. **D_2O is naturally present in formation water (~150 ppm)**, but any additional presence measured in the Dean-Stark-extracted water indicates filtrate contamination. For this pilot, a D_2O concentration of ~600 ppm was used in the coring fluid, ensuring high analytical accuracy in detecting invasion and correcting saturation estimates.

The pilot involved six coring rounds each retrieving ~40 ft of core. Cores were sectioned into 3-ft intervals, with start and end depths clearly marked. Fluids surrounding each core section were collected in labelled plastic bottles for subsequent lab analysis (Fig. 2). After retrieval, cores were stored vertically in sealed containers, allowing natural separation of oil and water to continue during transport.

In the laboratory, collected fluids were separated from mud using centrifugation. The volumes of oil and water recovered from each core section were added back to Dean-Stark-derived saturations, significantly improving the material balance. These values were then corrected to in-situ conditions using the formation volume factor for oil and stress correction factors for compaction effects on pore volume.

Figure 2—2(a) through 2(e) displaying Sequential workflow of Liquid Trapper core handling after recovery at surface. Once retrieved, the cores are placed on an iron mesh platform and cut at every 3-ft tubing joint. Fluids released from the tube during cutting are drained and collected in plastic bottles. When the core "half-moon" sleeves are opened, gases escape and additional fluids are expelled; these are combined with the earlier collected fluids. The core sleeves are then re-assembled, and the top and bottom depths are marked with lines and values. To preserve core integrity, the samples are coated with wax to seal the outer surface, wrapped in plastic covers, and properly labelled. Finally, the wax-coated cores are packed in wooden boxes with foam padding to fill gaps, ensuring safe protection during transportation and storage at the company's core facility.

Laboratory core experiments

In this study, two types of core acquisition strategies were adopted based on the objectives for each reservoir interval. Conventional cores were collected from the upper layers of the reservoir, primarily for conducting

Routine Core Analysis (RCA), petrography, and sedimentology studies. In contrast, Liquid Trapper cores were acquired from the lower reservoir layers, which represent the main EOR target zones.

While RCA was also performed on the Liquid Trapper cores, the primary focus was on estimating Remaining Oil Saturation (ROS) in partially swept and unswept zones, and Residual Oil Saturation (Sor) in fully swept zones. A detailed and flexible laboratory workflow was developed to meet these objectives and executed in three progressive batches.

- Batch 1 served as a pilot to test the initial workflow on a limited number of selected samples.
- Batch 2 continued the workflow only if Batch 1 results were successful; otherwise, modifications were made and applied to Batch 2.
- Batch Batch 3 followed a similar approach: it adopted the validated workflow if Batch 2 was successful, or incorporated further improvements if needed.

This staged approach allowed for optimization of lab time and cost, while ensuring that saturation estimates were obtained with reasonable accuracy.

Sample selection for the batches was guided by integrating high-resolution CT scan imagery and log data to identify homogeneous intervals, which improved the reliability of plug and whole core selection. In Batches 1 and 2, only a limited number of samples were analysed, specifically from partially and fully swept zones, to validate the workflow. For Batch 3, sampling was done at 1-foot intervals to achieve high-resolution coverage of the entire target zone. Additionally, selected core sections were preserved for future studies, including potential native-state core experiments.

Batch-1 workflow and results

In batch-1, 13 full diameter (FD) samples, 2 plug samples, and 15 liquid trapper mud samples were selected for lab experiments. FD samples allow accurate mass balance by reconciling Dean-Stark brine volumes with expelled fluids but are unsuitable for fractured sections. Plug samples provide denser saturation data and enable independent oil and water volume measurements via Dean-Stark and NMR, though their results cannot be directly reconciled with expelled fluids.

Fig. 3 illustrates the visual workflow followed in batch-1, which includes measurements such as porosity, permeability (ambient and at 3,900 psi), grain volume, pore volume, grain density, fluid saturation by Dean-Stark, deuterium concentration, invaded fluid, and corrected fluid saturation.

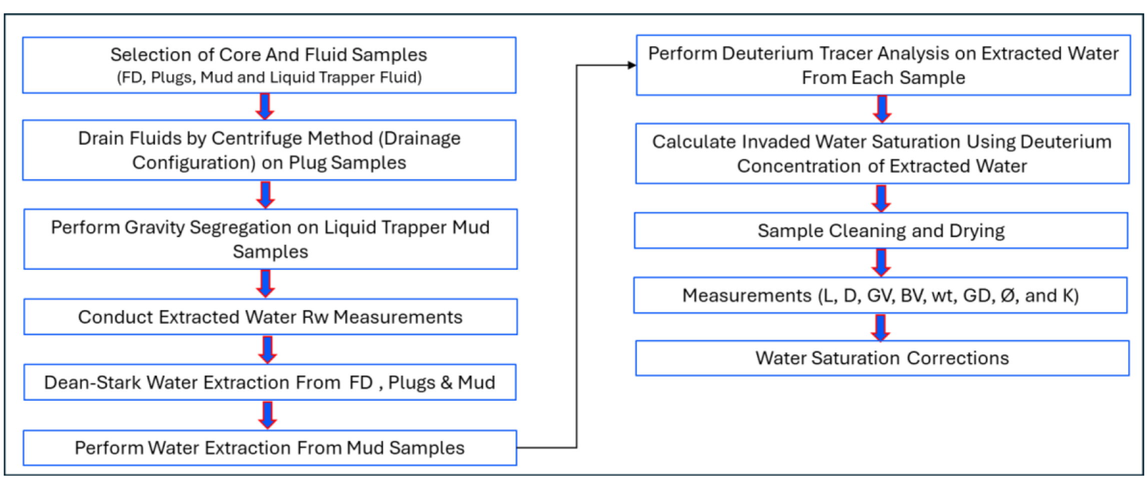


Figure 3—Displaying visual workflow for Batch-1 analysis. The process begins with core plug selection using CT scan images and progresses through key laboratory steps, primarily Dean-Stark water extraction followed by corrections using deuterium tracer and invaded-water saturation data. The workflow concludes with the determination of final water saturation (Sw).

Before loading samples into the Dean-Stark apparatus, pore water extraction by centrifuge was carried out on the plug samples to obtain water for formation water resistivity (Rw) measurements. The extracted water was analysed using a microresistivity cell at ambient temperature.

Subsequently, all full-diameter (FD) and plug samples were processed through Dean-Stark extraction to determine fluid saturations. The water collected from this process was also used for deuterium tracer analysis. Additionally, 15 mud samples and formation water were loaded into the Dean-Stark unit for water extraction and tracer testing.

Fluids recovered from the Liquid Trapper compartments were analysed separately. The mud samples, stored in bottles from the trapper system, were initially weighed, poured into 1L funnels, and allowed to settle under gravity. This gravity segregation process revealed minor oil traces (less than 0.05 cm3) in a few samples, indicating limited oil capture during coring.

The two horizontal plug samples were also run through a centrifuge before Dean-Stark loading. No oil was observed during centrifugation, and 1.00 cm^3 of water was produced, which was used for further Rw and pore saturation analysis. The deuterium concentration of the extracted water and corresponding mud samples was used to estimate invaded water saturation (Fig. 4a). The average deuterium concentration from mud samples was applied to interpret filtrate invasion across all sampled core sections.

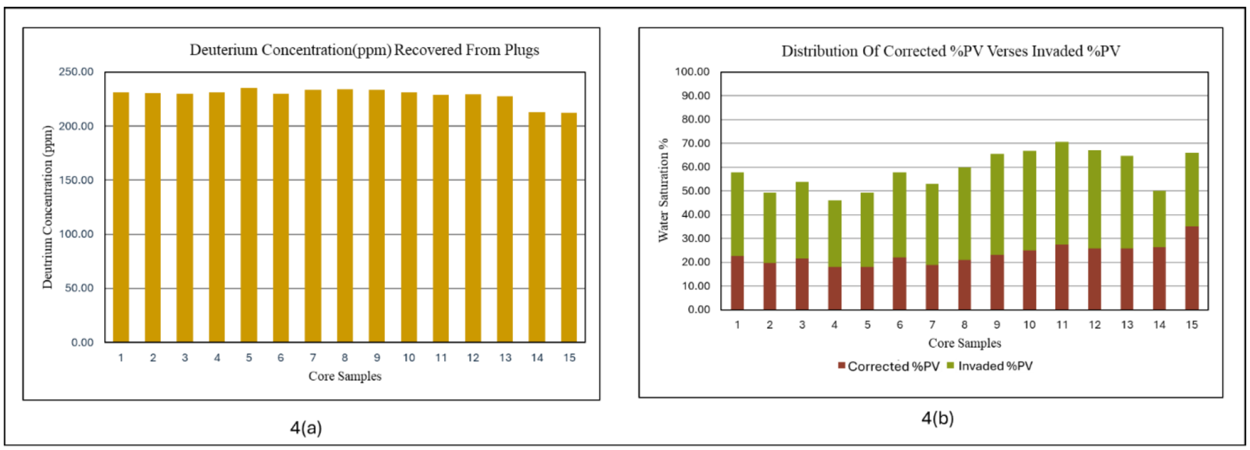


Figure 4—(a) Histogram showing total deuterium (D_2O) concentrations measured in water extracted during the Dean-Stark experiment. Elevated deuterium levels above the baseline formation water threshold of 150 ppm indicate the extent of mud filtrate invasion into the core plug. 4(b) Stacked histogram representing the volume of water corrected based on deuterium concentration measurements alongside the volume of water extracted from mud samples, expressed as a percentage of pore volume. The measured water volumes range between 45% and 70% of the pore volume

Following saturation analysis, all plug samples underwent cleaning procedures—first in hot refluxing toluene to remove hydrocarbons, then in hot methanol to dissolve salts. Samples were then oven-dried at 100°C to constant weight, cooled to room temperature, and subsequently analysed for porosity and permeability. The final step included correcting total water saturation for pore volume and mud invasion, thereby completing the Batch-1 analysis (Fig. 4b)

Batch-1 results (Baseline Data Quality, Limited ROS Reliability)

Fig. 5(a) presents the distribution of final oil saturation (So) and total water saturation (Sw) measured during Batch-1. Neither the full- diameter (FD) nor plug data achieved reliable Remaining Oil Saturation (ROS) estimates. Batch-1 results highlighted fundamental issues of reliability in core-based saturation measurements. **Material balance (So + Sw) averaged only ~0.77 (range: 0.71-0.83) versus the expected ~1.0, indicating 20-30% under-accounting of pore fluids**. Such deficits suggest systematic measurement errors (Dean-Stark limitations, fluid losses during retrieval, porosity measurement uncertainties). Although general trends in saturations matched log responses Fig. 5(b), deviations of up to 30 saturation units were observed. **In terms of EOR decision-making**, these results underscore the high uncertainty in ROS determination, making it difficult to establish a reliable baseline for miscible gas injection design. Batch-1 therefore set the stage for methodological refinement but offered limited quantitative input for pilot evaluation and decision making.

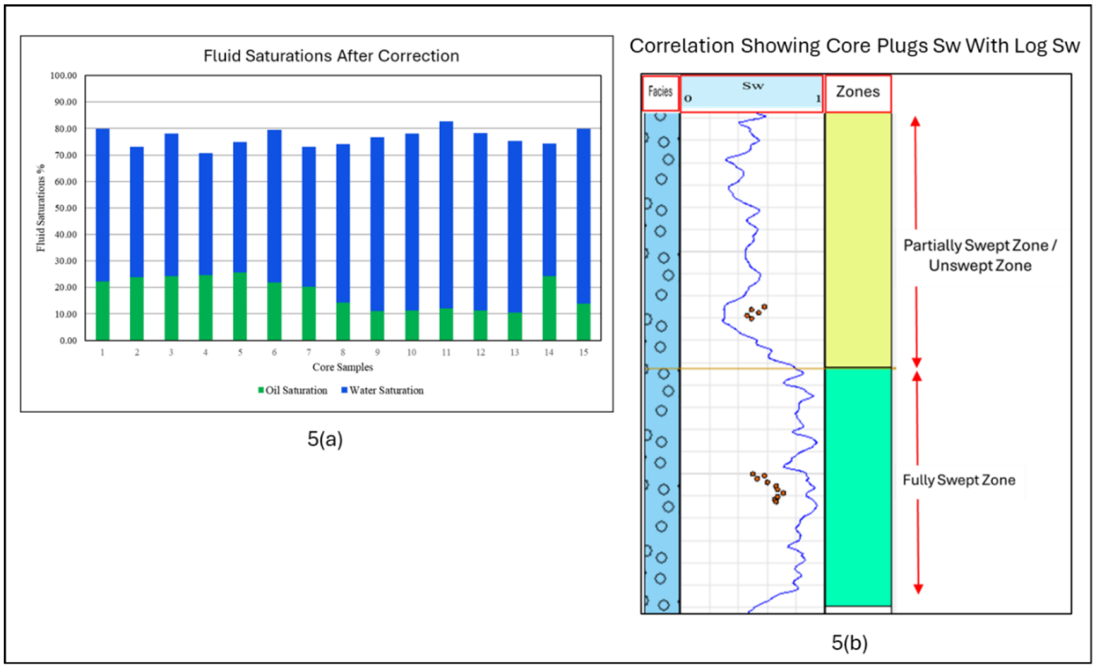


Figure 5—(a) Stacked histogram presenting corrected oil and water saturations measured in Batch-1. The total fluid saturation sums to less than 81%, indicating that full material balance close to 100% was not achieved. This discrepancy may result from calculation errors, fluid loss during core handling, or other uncertainties. 5(b) Correlation of core plug water saturation measurements (red dots) with water saturation log data. While the plug saturations generally follow the log trend, deviations of ±15 porosity units occur—positive deviations in the partially swept zone and unexpected negative deviations in the fully swept zone.

The next phase, Batch-2, is focused on evaluating whether core plug analysis provides improved saturation estimates compared to whole-core measurements.

Batch-2 workflow and results

The primary objective of Batch-2 was to reduce the uncertainty observed in **Residual Oil Saturation (Sor)** estimates from Batch-1. To achieve this, two sets of core plugs were selected from intervals adjacent to those investigated in Batch-1. These represent two key displacement conditions:

- A fully swept zone, where the expected water saturation (Sw) is ≥ 0.8-0.9
- A partially swept zone, where Sw is expected to be < 0.4

Core plug locations were selected using CT scan and log data, focusing on consolidated and competent intervals to maximize the quality and reliability of samples. The selection aimed to capture Sw values as close as possible to the maximum saturation in the swept zone and the minimum saturation in the unswept zone. To minimize disruption, the number of core sections opened was limited to one per zone (i.e., one for swept and one for unswept), avoiding sample locations that would require opening more than two Liquid Trapper (LT) core sections

Fig. 6 outlines the experimental workflow adopted in Batch-2, which includes the following stages:

- NMR measurements under ambient conditions (in three saturation states)

- Core flooding to establish residual oil saturation (Sor)
- Dean-Stark extraction for volumetric and gravimetric fluid analysis
- Routine Core Analysis (RCA) to determine porosity and permeability

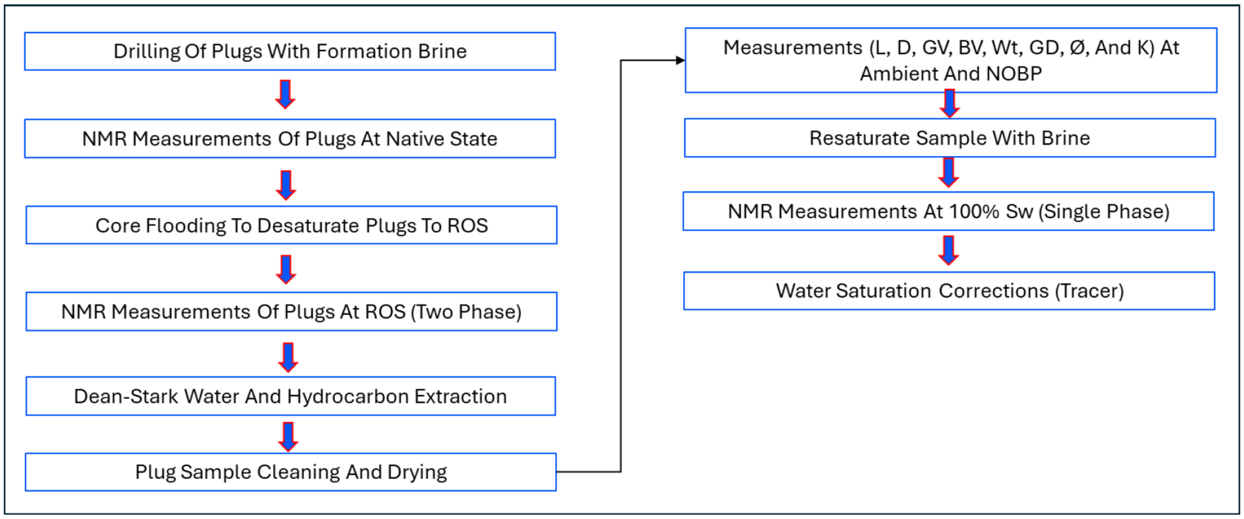


Figure 6—Displaying visual workflow for Batch-2 analysis. The process begins with core plug selection using CT scan images and drilling using formation brine, progresses through key laboratory steps, primarily NMR measurements, Dean-Stark water extraction followed by corrections using deuterium tracer and invaded-water saturation data. The workflow concludes with the determination of final water saturation (Sw).

During sample preparation core plugs (1.5-inch diameter, 5 cm length) were cut using formation brine to prevent fluid invasion. Plugs were trimmed to exclude mud-invaded zones; trimmed ends were marked and preserved for future studies. NMR was used to complement Dean-Stark volumetric and gravimetric data by providing accurate oil and water saturation estimates through three stages:

- Native state NMR: Immediately after plug cutting
- Two-phase NMR: After core flooding, on air/gas-free plugs containing immobile oil and mobile water
- Single-phase NMR: After Dean-Stark and RCA, on plugs re-saturated to 100% water saturation

Following native-state NMR, core flooding was performed using formation brine to reach residual oil saturation (Sor). This process helps expel any remaining gas/air and simulate field saturation conditions. Flooding was conducted in multiple cycles, with and without backpressure, to ensure full saturation and validate the procedure by comparing measured permeabilities. The Two-Phase NMR conducted after this stage provides a reference to calibrate native-state NMR, assuming the sample now contains only immobile oil and mobile water.

After Two-Phase NMR, Dean-Stark extraction was used to recover oil and water from the pore space, followed by Routine Core Analysis to determine porosity and permeability. The final step involved Single-Phase NMR on the same samples, now fully saturated with water (Sw = 100%), to calibrate the earlier NMR measurements and provide a robust end-to-end assessment of fluid saturations and rock properties (Fig. 7).

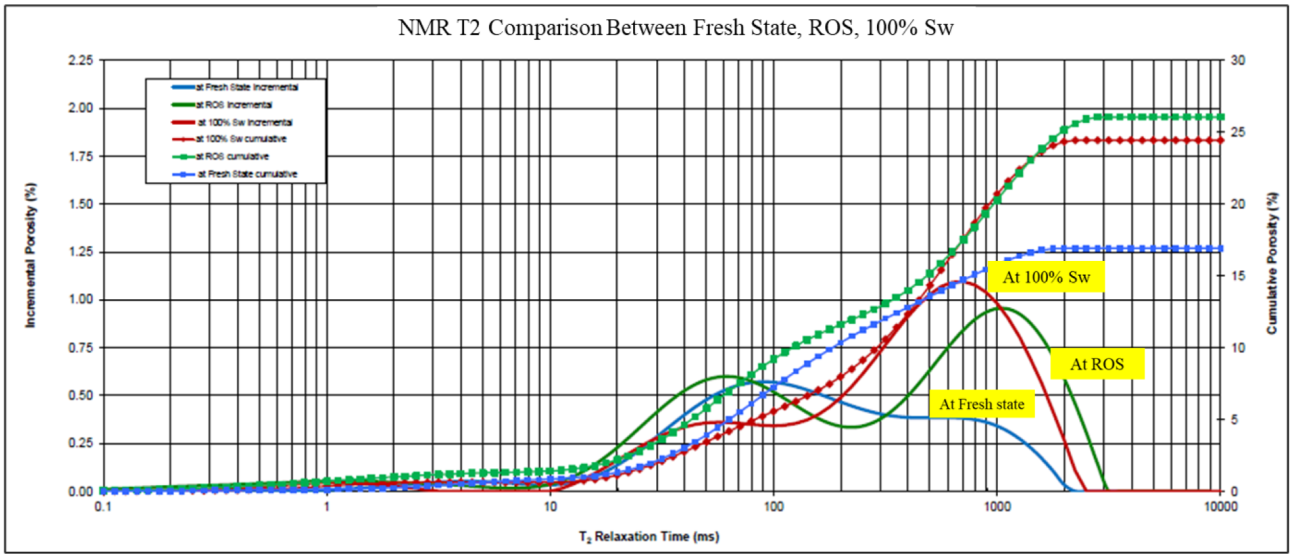


Figure 7—Comparison of NMR T2 distributions measured in the Native state, Remaining Oil Saturation (ROS), and plug saturated at 100% water saturation (Sw). The T2 distribution at ROS exhibits longer relaxation times compared to the 100% Sw plug, indicating the presence of residual oil. Native-state T2 peaks are smaller due to a lower hydrogen index (HI) volume, reflecting a mixture of pore fluids including oil, water, and possibly gas. Additionally, mud invasion during coring may reduce the overall NMR signal intensity in the native state.

Batch-2 results (Improved Saturation Accounting but Questionable ROS Detection)

Fig. 8 presents the outcomes of the Batch-2 analysis, demonstrating a significant improvement in data quality compared to Batch-1. The Dean-Stark **material balance (So + Sw) improved from 0.60-0.70 in Batch-1 to greater than 0.95 in Batch-2** (Fig. 8), indicating a more complete accounting of pore fluids and improved reliability of the measurements.

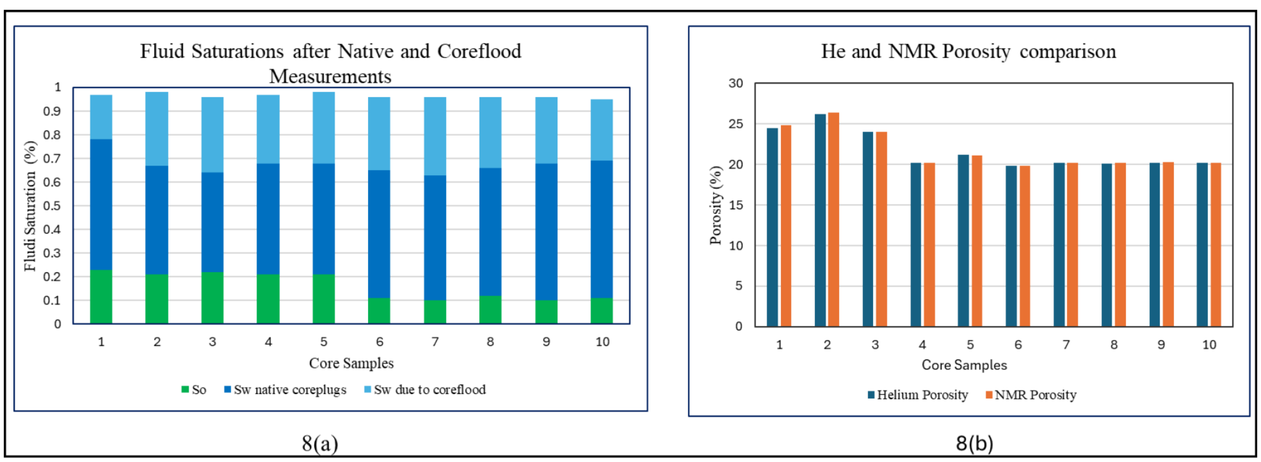


Figure 8—a Stacked histogram showing fluid saturations measured in the native state and post-coreflooding. The total fluid saturation sums to approximately 95%, indicating near-complete material balance—an improvement compared to Batch-1 results. 8(b) Comparison of helium porosity and NMR porosity histograms for all core plugs. The close match between helium and NMR porosity validates the reliability of the NMR measurements and the derived fluid saturations.

Porosity estimates obtained through volumetric, gravimetric, and NMR techniques were largely consistent (Fig. 8), further supporting the quality of the measurements. These results confirm the earlier hypothesis from Batch-1: that a significant portion (approximately 20-30%) of the pore volume was initially filled with gas rather than fluids, likely due to loss of mobile oil during coring, retrieval, or surface handling. The Batch-2 findings support this assumption, highlighting that some of the mobile oil was expelled before measurements could be made.

During the core flooding experiments, **no measurable oil production was observed**. This was expected for plugs sampled from the lower (fully swept) reservoir zone, where most mobile oil would have already been displaced during reservoir production. However, the same outcome was observed in plugs from the upper reservoir zone, which were expected to be unswept or partially swept and should have retained mobile oil. This unexpected lack of oil production raised concerns about the effectiveness of the Batch-2 workflow in accurately capturing remaining oil saturation (ROS) in these zones.

Despite this, comparison with log-derived saturations shows a good overall match (Fig. 9), suggesting that the core data while affected by fluid losses still reflects general reservoir trends. However, it is important to account for the potential loss of mobile water during coring and handling when correlating core data with logs.

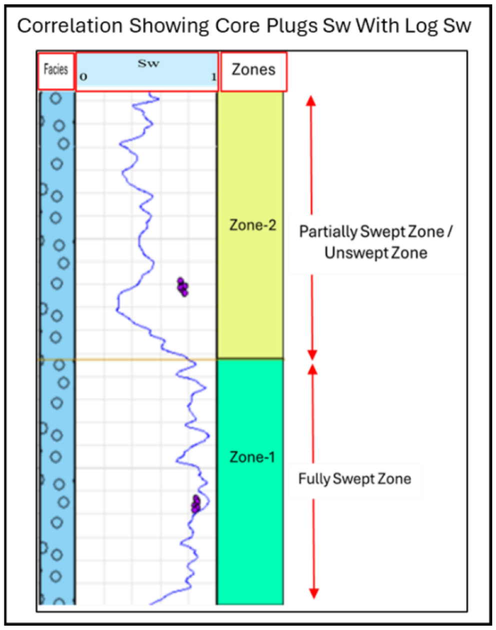


Figure 9—Correlation of core plug water saturation measurements (red dots) with water saturation log data from Batch-2 analysis. The plug saturations generally follow the log trend, with deviations of up to +20 porosity units observed in the partially swept zone, indicating potential fluid loss, measurement uncertainties, or the need for improved saturation estimation. In contrast, a strong match between plug and log saturations is observed in the fully swept zone, representing an improvement over Batch-1 results in the same area.

For EOR pilot planning, Batch-2 suggests that while saturation trends can be trusted for **static reservoir modeling**, the lack of measurable ROS recovery introduces uncertainty in **estimating incremental recovery potential from gas** injection projects.

Batch-3 workflow and results

Based on insights from Batch-2, using core plugs combined with core flooding and NMR measurements improved confidence in the Dean-Stark method. Since the main goal is to determine remaining oil saturation (ROS), an additional centrifuge step was introduced to better estimate residual oil saturation (Sor). Some Batch-3 plugs were cut from preserved liquid trapper cores.

Batch-3 plugs were split into two groups: one for ROS and Sor measurement using core flood and centrifuge only (to reduce handling), and the other also includes NMR for independent oil saturation measurement. Fig. 10 illustrates this batch-3 workflow, which includes core flooding to establish ROS, centrifuge water displacement, NMR measurements, Dean-Stark fluid analysis, and routine core analysis.

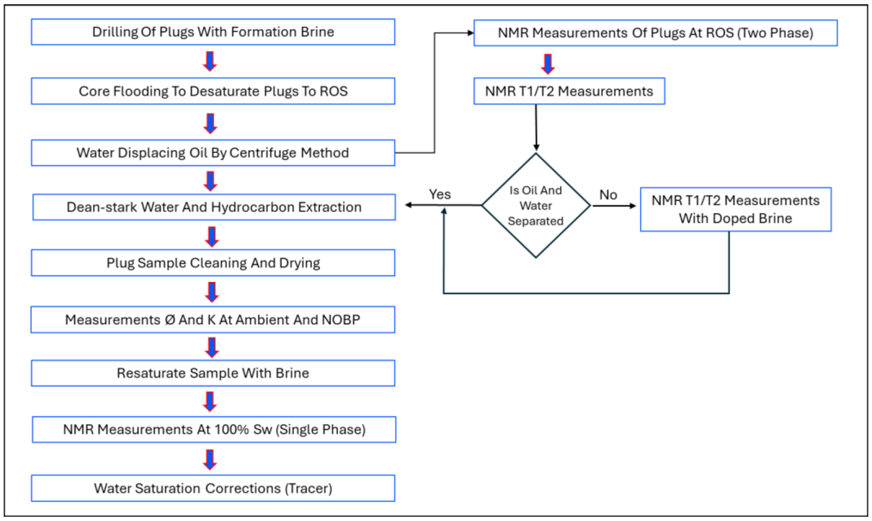


Figure 10—Displaying visual workflow for Batch-2 analysis. The process begins with core plug selection using CT scan images and drilling using formation brine, progresses through key laboratory steps, primarily NMR measurements, Dean-Stark water extraction followed by corrections using deuterium tracer and invaded-water saturation data. The workflow concludes with the determination of final water saturation (Sw).

Using this workflow, plugs are first drilled with formation brine and desaturated via core flooding to reach ROS. Then, centrifugation at rates based on permeability continues until production slows to a stable equilibrium. Plugs from the centrifuge batch undergo Dean- Stark analysis for fluid saturation verification, while those in the NMR batch underwent further NMR T1/T2 measurements. If NMR shows oil-water separation, plugs proceed to Dean-Stark analysis; if not, doped synthetic brine (with $MnCl_2$) is injected via core flooding before repeating NMR and Dean-Stark analyses. Finally, all plugs undergo routine core analysis, completing the experiment.

Batch-3 results (Validated ROS Measurements and Enhanced NMR Diagnostics)

The results from Batch-3 are encouraging, meeting key experimental objectives and improving confidence in the data quality. First, material balance of fluid saturations was successfully achieved in the majority of samples (Fig. 11a). However, a few samples exhibit pore volume deficits that is, the measured fluid saturations do not fully account for the pore space (Fig. 11b). This suggests a loss of mobile oil during coring, handling, or related operations, consistent with observations from previous batches.

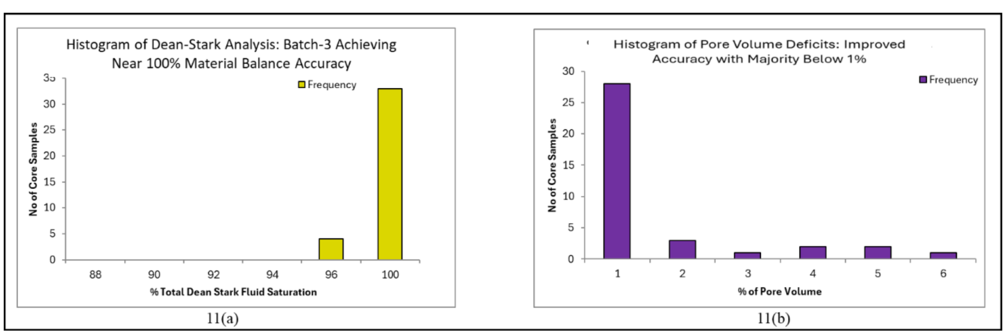
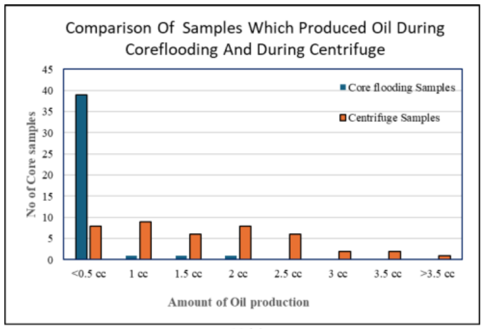


Figure 11—(a) displayed a histogram of the total number of samples analyzed using the Dean-Stark method, with material balance achieving nearly 100% accuracy. Batch-3 showed significant improvement, with all samples exceeding 95% material balance and most nearing 100% accuracy. Fig. 11(b) presented a histogram of pore volume deficits, indicating that the majority of samples had less than 1% unaccounted volume—likely due to gas presence or minor fluid loss—reflecting more realistic results. However, a few samples still exhibited pore volume deficits greater than 1%, which marked a significant improvement compared to the results observed in Batch-1 and Batch-2 analyses

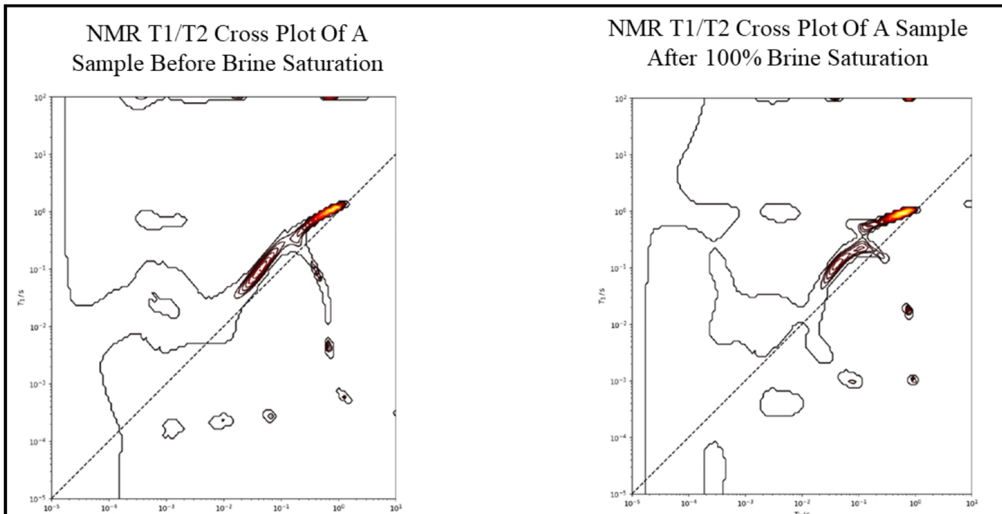


11(c)

Figure 11c—Histogram comparing oil volumes measured during core flooding and centrifuge tests. The centrifuge results reveal additional movable oil in the core plugs that core flooding was unable to recover, likely due to limitations in overcoming capillary forces and the presence of oil in tight pore spaces. These findings indicate that core flooding alone is insufficient for accurately quantifying fluid saturations in laboratory experiments and must be complemented with centrifuge testing.

The second major objective was measuring Residual Oil Saturation (Sor) with reasonable accuracy was also largely met. During core flooding, small volumes of oil were recovered from most plugs, indicating that some mobile oil remained. Interestingly, subsequent centrifuge tests on these same samples yielded additional oil, implying that flooding alone may not fully displace all mobile hydrocarbons. This highlights the continued need for centrifuge-based methods to complement core flooding and improve ROS determination (Fig. 11b and 11c).

To further characterize fluid saturations, T1/T2 NMR experiments were conducted after the centrifuge process. The initial T1-T2 measurements on two test samples at ROS showed partial separation between oil and water signals. However, the inverted spectra lacked clear relaxation contrast, making quantification of oil and water saturations unreliable. This is illustrated in Fig. 12, which compares T1-T2 spectra for one sample before and after re-saturation to 100% water. Although some spectral separation was observed, it was insufficient for accurate fluid differentiation.



12(a) 12(b)

Figure 12—(a) NMR T1/T2 cross-plot of the core before brine saturation, showing minimal surface relaxation effects and poor contour separation between bound water, movable water, and hydrocarbons. 12(b) After saturating the core with 100% formation brine, surface relaxation effects dominate due to strong brine-clay interactions, resulting in clear contour separation in the NMR T1/T2 plot.

To improve peak resolution, doped brine was introduced in the remaining samples. This modification enhanced the NMR response, enabling better separation between water and hydrocarbon signals.

NMR T2 measurements were then repeated on these samples both before and after flooding with doped brine to assess the improvement in fluid phase differentiation. The post-flooding NMR data exhibited clear and distinct signals for water and hydrocarbon phases, confirming the effectiveness of the doped brine approach.

Additionally, the cumulative volumes derived from NMR T2 distributions (post-core flooding and post-cleaning), along with saturation- derived pore volumes and helium-measured pore volumes, were generally consistent across all samples. Notably, the doped-brine test results showed slightly higher pore volumes compared to the other NMR methods, as illustrated in Fig. 13. This discrepancy may reflect enhanced signal recovery or improved wetting of the pore system due to the brine doping

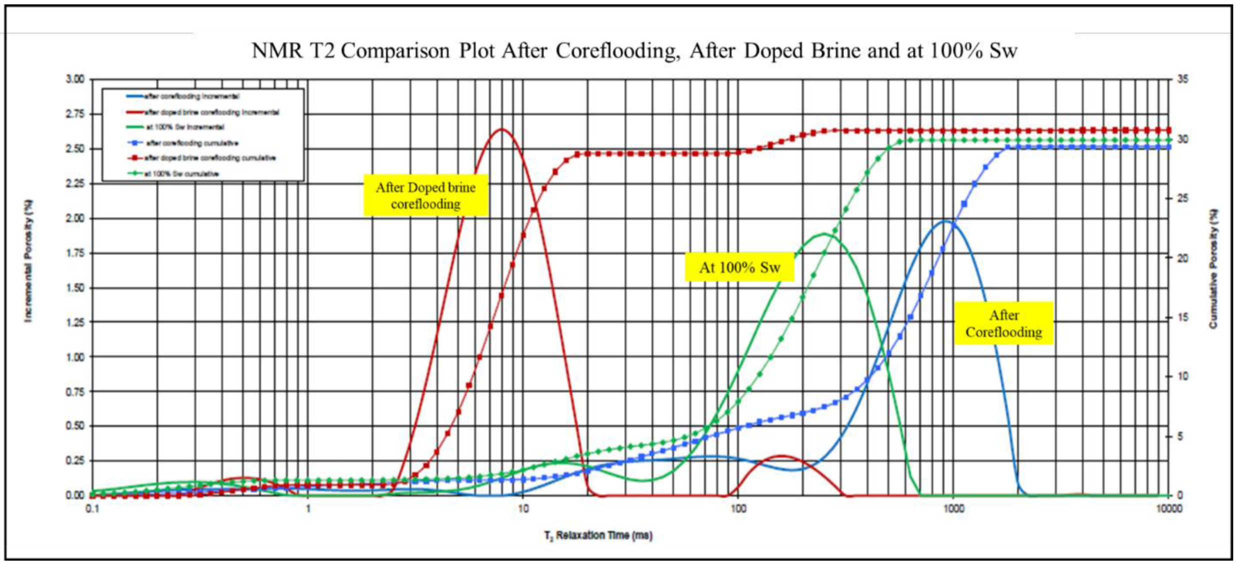


Figure 13—NMR T2 distributions comparing core flooding with doped brine versus normal core flooding at 100% formation brine saturation. At 100% formation water saturation, NMR T_2 shows large free-water peaks in large pores and smaller peaks for bound/capillary water in small pores. Using doped brine shifts water T_2 to shorter times, clearly separating hydrocarbon (oil) signals at longer T_2. Core flooding with normal formation brine also separates oil from bound water, but the distinction is less pronounced than with doped brine.

For EOR design decisions, Batch-3 provides three critical insights:

- **Residual Oil Presence Confirmed** - The measurable oil recovery validates that target zones still retain producible hydrocarbons, supporting the case for miscible/immiscible gas injection pilots.
- **Injection Strategy Guidance** - The incremental oil recovered via centrifuge (beyond flooding) highlights that certain pore systems require higher displacement energy. This implies that miscible gas injection with high-pressure gradients may be more effective than simple waterflood continuation.
- **Pilot Calibration Data** - Reliable ROS values and improved NMR-based fluid characterization mean pilot simulation models can now be anchored with realistic parameters, improving forecasts of recovery efficiency and breakthrough timing.

The following table shows comparative summary of all the three batch results:

Table 1—Comparative Summary of Batch-1, Batch-2, and Batch-3 Results

Batch	Material Balance (So+Sw)	Oil Recovery (Flooding/Centrifuge)	Data Reliability / Key Issues
Batch-1	~71-83% (avg~77%) → *20-30%* fluid deficit	None observed (ROS not measurable)	Low reliability-systematic errors suspected (Dean-Stark limitations, fluid losses during coring/handling, porosity measurement uncertainties). Only qualitative consistency with log trends.
Batch-2	Improved to >95% (vs. 60-70% in early Batch-1 subsets)	No measurable oil recovery during flooding (even in partially swept zones)	Moderate reliability-balances improved and porosity consistent across methods; however, absence of oil production suggests underestimation of ROS due to fluid losses during retrieval/handling.
Batch-3	Majority of samples achieved near 100% closure (minor pore volume deficits in few cases)	Oil recovered by flooding; additional oil recovered via centrifuge → confirms ROS presence	High reliability-validated ROS measurement; NMR with doped brine enabled clear water/oil differentiation. Provides actionable inputs for EOR pilot calibration and injection design.

Conclusions

- EOR pilot projects, like the miscible gas flood in the Minagish field, are vital for assessing commercial viability in heterogeneous reservoirs.
- Accurate estimation of Remaining Oil Saturation (ROS) and Residual Oil Saturation (Sor) in unswept, partially swept, and fully swept zones is critical for evaluating EOR effectiveness.
- Conventional coring struggled to preserve in-situ fluid saturations due to fluid loss during retrieval, resulting in unreliable ROS data.
- The Liquid Trapper coring method, enhanced with deuterium tracer in the mud, improved fluid retention and reduced uncertainty in saturation measurements.
- A combined approach—using conventional coring in upper layers and Liquid Trapper coring in lower EOR-targeted zones— balanced cost and data quality.
- Laboratory workflows executed progressively in three batches allowed optimization of sample selection, analytical methods, and data accuracy.
- Integration of CT scans and log data helped select homogeneous intervals for reliable whole core and plug samples.
- Batch-1 revealed significant fluid loss during coring, indicated by poor material balance and underestimated oil saturations.
- Batch-2, focusing on core plugs and improved analytical techniques (NMR, core flooding, Dean-Stark), greatly enhanced material balance and saturation accuracy.
- Batch-3 introduced centrifuge steps and doped brine during NMR, further refining ROS and Sor measurements and improving separation of hydrocarbon and water signals.
- NMR measurements combined with doped brine demonstrated clear phase differentiation and consistent pore volume estimates, advancing fluid saturation characterization.
- Despite some residual uncertainties from core handling and fluid invasion, the study's integrated approach provides a robust framework to reduce saturation estimation errors and support effective EOR design in complex carbonate reservoirs.

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